

Figure 1: The basic wireless network

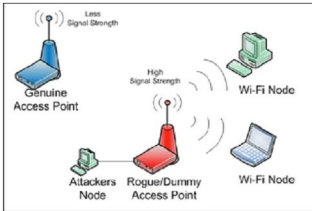


Figure 2: A dummy AP attack

Unfortunately, these security mechanisms are either flawed by design, or are not adequate for IT infrastructures where data carried on wireless channels is sensitive.

With this basic understanding of Wi-Fi security, let us now discuss a few security attacks. In general, there are four categories of possible attacks.

Passive attacks: In this type, the attacker listens or eavesdrops on an open wireless channel by using a wireless modem rigged to work in a promiscuous mode. All traffic packets that contain important information, such as MAC addresses, packet sequences, etc, are stored. Passive attacks may not necessarily be malicious in nature, but help provide information for active attacks. Since passive attacks take place silently, they are almost impossible to detect and stop. Attackers using passive methods usually capture and store data, and use a packet-deciphering tool to decrypt it and steal information. This is especially true in case of the WEP protocol, due to its inherent lack of security. Passive attacks are also called wireless war-driving.

Active attacks: Once an attacker gets sufficient information by passive attacks, an active attack can be tried. Common examples are denial of service, IP spoofing, etc. In case of spoofing, the attacker gains access to an unauthorised wireless station, and performs packet crafting to impersonate a valid and authorised station. Wireless nodes are incapable of detecting this, and end up connecting to the attacker's station and revealing

information. By extending this technique, the attacker can now plant a denial of service attack on a particular node in order to disrupt its services. Typically, a SYN flood method is used, because it is sufficient to generate a packet storm on the given wireless connection bandwidth.

MITM attacks: We did explore man in the middle (MITM) attacks in one of the previous articles and most of that applies to wireless networks as well. The only technical difference here is that the attacker gains information of an actively used SSID of an AP, instead of an on-the-wire session. As shown in Figure 2, a dummy AP with exactly the same name is created by the attacker, and the signal power is raised to such an extent that the nodes are fooled into believing that it is the AP they should connect to. This creates an MITM situation. These dummy APs, also called rogue points, are usually set up close to the nodes to be hacked.

Signal-jamming attacks: Unlike the above techniques, this method uses wireless radio transmission techniques to create an attack. In this type, the attacker uses a powerful antenna and a signal generator, and creates frequency patterns in the same range as wireless signals. The frequency patterns are modulated with powerful radio frequency ripples, to create a wireless signal storm. This results in the jamming of the APs as well as the nodes, thus disabling their connectivity. While such an attack was just a theory previously, with a growing number of wireless networks these attacks have now occurred more often than earlier.

Besides these, there are a few other types of attacks, some of which make use of one or more of the attacks mentioned above.

802.11 injection attacks: Modern attackers tend to go deep into the protocol stack in order to plant an attack. For wireless networks, an attacker can first perform a passive attack to understand the protocol frame structure, and then create 802.11 protocol datagram frames and insert those into the network. This is usually done either to create a false packet stream as a hindrance for a wireless network, or to sniff the network further in an active mode. The response 802.11 frames are then captured again, interpreted and modified to perform an MITM attack. Since this attack happens at Layer 2, it is very tough to detect.

Wireless packet injection: Here, passive attacks are used to capture traffic, which is then analysed. However, there can be situations in which there won't be enough traffic to generate sufficient data, which can lead to time-consuming or futile hacking efforts. Hence, attackers use wireless packet injection techniques whereby, besides the 802.11 frames, IP datagrams are sent to the target AP. Though the AP will drop such packets as unauthorised, this gives the attacker the necessary amount of traffic, which is captured and fed into key-cracking utilities. Since the attacker controls the packet-generator utility, specific data patterns are intentionally created to map the AP's behaviour in terms of response packets, which further helps in reducing the cracking time.